

Optimal design and energy management of a hybrid PV-Wind system with hydrogen and gravity energy storage: An off-grid sustainable alternative for coal power in Morocco

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ABSTRACT

Energy storage integration is vital for reliable power supply as reliance on renewables grows. This study investigates the co-optimization and control of an off-grid hybrid system—comprising photovoltaics (PV), wind turbines (WT), hydrogen storage, and gravity energy storage (GES)—as a sustainable alternative to a 624 MW ultra-supercritical coal unit in Morocco. Unlike prior work, this paper explicitly quantifies the distinct roles of GES and hydrogen in coal plant replacement scenarios. A unified framework is proposed to size all system components while performing 8760-hour dispatch simulations to ensure uninterrupted power supply, achieving a 0% loss of power supply probability (LPSP). At this reliability level, the optimal configuration includes ~1000 PV modules, 594 wind turbines, a GES unit (5 m diameter, 714 m height), and substantial hydrogen infrastructure: a 790 MW electrolyzer, 650 MW fuel cell (FC), and 260 t storage tank. The resulting levelized cost of electricity (LCOE) is 0.23 €/kWh. The system reliably meets demand by leveraging PV, WT, FC, and GES. Intermittency in PV and wind is mitigated through the complementary roles of hydrogen and GES. Hydrogen production aligns with renewable generation, while GES exhibits frequent deep-cycling, highlighting its key balancing function. This analysis demonstrates that a well-sized and controlled PV–WT–Hydrogen–GES system can serve as a credible, clean alternative to coal-based generation. It underscores the potential of hybrid energy storage systems in enabling sustainable, off-grid power solutions, particularly in regions with abundant renewable energy resources.

1. Introduction

The global energy sector stands at a pivotal juncture, compelled by the dual imperatives of climate action and sustainable development to phase out fossil fuel dependency and accelerate the adoption of renewable energy systems [1]. Coal-fired power generation, despite incremental efficiency improvements in ultra-supercritical (USC) technologies, remains a leading contributor to greenhouse gas (GHG) emissions, accounting for over 30% of global CO₂ emissions in 2023 [2,3]. While USC plants achieve thermal efficiencies of up to 45%, their emissions intensity—approximately 700–750 kg CO₂/MWh—makes them

incompatible with the Paris Agreement's 1.5°C pathway [4].

Morocco's renewable energy ambitions are outlined in its National Energy Strategy 2030, targeting 52% renewable electricity capacity and a 45.5% reduction in GHG emissions by 2030 [5]. However, the intermittent nature of solar and wind resources poses significant challenges to grid reliability and requires the optimal utilization of energy storage systems (ESS). These systems would ensure both diurnal and seasonal arbitrage. While lithium-ion batteries dominate short-duration storage, their high costs and degradation rates limit scalability [6]. Other emerging storage systems have gained increasing interest, among them hydrogen storage and gravity energy storage.

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Nomenclature			
C	Cost (\$)	I_f	Predicted direct irradiance (W/m^2)
C_{cap-i}	Capital expenditures (\$)	k	Temperature coefficient
Cap_{PV}	PV Capital cost (\$)	m	Mass (kg)
Cap_{WT}	Wind turbine Capital cost (\$)	M_{tank}	The quantity of stored hydrogen (kg)
Cap_{GES}	GES Capital cost (\$)	N_{pv}	number of PVs
Cap_{ELz}	Electrolyser Capital cost (\$)	N_{WT}	Number of wind turbines
Cap_{FC}	Fuel cell Capital cost (\$)	P	Power (kW)
$Cap_{O\&M-PV}$	PV operation and maintenance costs (\$)	P_{ch}	Power stored (kW)
$Cap_{O\&M-WT}$	Wind turbine operation and maintenance costs (\$)	P_{disch}	Power discharged (kW)
$Cap_{O\&M-GES}$	GES operation and maintenance costs (\$)	P_{H2}	Produced Hydrogen (kg)
$Cap_{O\&M-ELz}$	Electrolyser operation and maintenance costs (\$)	P_{ELz}	Electrolyzer rated power
$Cap_{O\&M-FC}$	Fuel cell operation and maintenance costs (\$)	P_{PV}	PV Power (kW)
C_{REP-i}	Operation and maintenance costs	$P_{Wind-rate}$	Wind rated power (kW)
C_{REP-i}	Replacement cost (\$)	r	Discount rate (%)
C_{DISP-i}	Disposal Cost (\$)	SOC	State of charge (%)
DNI	Direct irradiance (W/m^2)	T	Temperature (k)
DOD	Depth of discharge (%)	T_{STC}	Temperature at standard test conditions (kt)
E	Energy (J)	V	Piston velocity (m/s)
$E_{battery(i)}$	Battery Capacity at a specific hour (kWh)	v	Wind speed (m/s)
E_{BAT}	Battery Capacity (kWh)	v_{cut-in}	Starting speed (m/s)
E_{tank}	Hydrogen tank's charge level.	$v_{cut-out}$	Stopping speed (m/s)
$E_{min-tank}$	Hydrogen tank's charge min level	V_{rated}	Rated wind speed (m/s)
$E_{max-tank}$	Hydrogen tank's charge max level	x	Variable vector
G_{STC}	Solar irradiance (W/m^2)	Greek symbols	
g	Gravitational acceleration (m/s^2)	σ	Discharge rate (%)
G_0	Solar constant (W/m^2)	η_{bat}	Efficiency (%)
G_f	Actual solar irradiance	η_{b-dich}	Discharge Efficiency (%)
h	Height (m)	η_{b-ch}	Discharge Efficiency (%)
H (t)	Hourly Humidity	ρ	Density

electricity, offers a compelling solution for long-term storage and sectoral decarbonization [7]. Recent advances in proton exchange membrane (PEM) electrolyzers, which achieve efficiencies of 70–80%, together with their projected cost reductions to \$700–1,000/kW by 2030, further enhance its viability [8].

Numerous scientific research publications have recently presented a number of applications of contemporary optimization algorithms and advanced software used for the optimization of the design and configuration of different hybrid renewable energy systems (RES) [9,10]. In [11] an energy coupling system that enables bidirectional electricity-hydrogen conversion, integrating renewable sources, has been studied for enhanced flexibility. Economic feasibility, environmental impact, and hydrogen production reliability have been considered. Authors in [12] explore electricity and hydrogen production from a hybrid offshore wind and floating PV system. A Novel Dung Beetle Optimization (DBO) algorithm was developed to optimize the hybrid system, with a detailed statistical analysis to confirm its robustness. Reference [13] propose a data-driven framework for optimizing the sizing of a hybrid PV/WT/H₂ energy system. In [14], an optimization process was conducted to determine the optimal sizing of key components, including PV, electrolyzer, hydrogen tank, electric boiler, and WT, ensuring efficient and sustainable energy management.

A solar PV system with hydrogen storage and battery system was proposed in [15]. It was optimized using several robust optimization scenarios before being contrasted with stochastic optimization, which relies on cumulative density functions. Due to its grid dependence, the PV arrangement has the lowest LCOE (269 €/MWh) and the lowest self-sufficiency (less than or equal to 30%) ratio, which eliminates the possibility of blackouts and power outages. In comparison to full-grid dependency, hybridizing solar PV with hydrogen/battery storage offers a more cost-effective alternative while lowering the likelihood of

blackouts and power outages.

In their proposed integrated energy system, authors of [16] used a Robust Design Optimisation (RDO), which takes into account the uncertainties in technical, economic, and environmental parameters, to optimise the configuration of the hydrogen refuelling system powered by photovoltaic and wind turbines and the fleet of diesel buses. Hydrogen-powered buses would replace 54% of the fleet of diesel-powered buses, lowering the LCOD standard deviation (SD) by 36%, the C.I. mean and SD values by 46%, and 51%, respectively. However, the LCOD mean would only increase slightly by 11%. These findings unequivocally demonstrate the system's potential for improved environmental and economic performance.

A dynamic forecasting model of a hybrid PV/GES for residential applications has been studied in [17,18]. The accuracy of the developed forecast model is assessed using a variety of performance indicators; prediction errors range from 13.45% to 23.16% for RMSE and from 4.06 to 11.27% for MAPE. The findings show that the suggested model's performance is within the range of other models. This suggests that, regarding the objective of predicting PV power generation, the model developed has promising accuracy and can be regarded as competitive among current models. Furthermore, for a number of uncertain parameters, authors in [19] carried out a comprehensive design optimization to determine the optimal levelized cost of hydrogen for a solar based electrolyser system. The region with the maximum yearly solar irradiation was found to have the lowest levelized cost of hydrogen (LCOH), with a levelized cost of 6.6 €/kg. According to the analysis, the greatest fluctuation in hydrogen unit cost was attributed to the discount rate uncertainty.

A comparative evaluation of four distinct meta-heuristic optimization techniques was presented in [20] to optimize a hybrid PV/WT with hydrogen and battery storage for three typical loads in a Cameroonian

location. The imperialist, cuckoo-search, artificial-bee-colony, and big-bang-big crunch algorithms are used to size six distinct configurations while minimizing the net present cost and enhancing dependability. The results showed that off-grid solar PV/wind/battery outperforms the other three algorithms in terms of cost and dependability. Reference [21] examined the most optimal approach to combine fuel cells, electrolyzers, and wind-solar PV to consistently satisfy power demands. A novel version of the Particle Swarm Optimization (PSO) method, subject to reliability requirements, minimizes system costs over a 20-year operating period. It takes into account the possibility that the three main components of the hybrid system including a PV, WT, and a DC-AC converter, might malfunction. The results show that outages have a direct effect on component dependability and cost, i.e., outages require a larger production system to provide the load with a satisfactory reliability level.

Although Zhen et al. have proposed an integrated PVWTH2 framework to cope with carbon trading uncertainties and optimize PVWTH2, they do not include a geometry-aware GES model or cooptimize PV-WT-H₂-GES at LPSP = 0%. [15]. Multiple 2024 studies enhance PV-WT-H₂ scheduling economics and equipment wear—e.g., multi-objective rolling optimization with alkaline electrolyzers to limit switching and idle time, and complementary WPH systems targeting economy + reliability; yet they omit GES coupling and coal-replacement reliability constraints [16]. In [17], a PV-WT-battery-FC sizing study for rural electrification refines hybrid sizing but does not address LPSP = 0% off-grid coal substitution or gravity storage [17]. Additional work on integrated wind-PV-H₂ at park scale and coordinated wind-solar-H₂-battery scheduling focuses on rolling dispatch and multi-timescale coordination rather than joint sizing + dispatch with GES [18]. An article in the 2024 Wiley Interdisciplinary Reviews outlines types and technoeconomics of energy storage, highlighting the need for methods for integrating renewables into the process, precisely the gap we fill via cooptimized PV-WT-H₂-GES with strict reliability. [19]. We also note emerging work on underground H₂ storage in hybrid systems, which our framework can incorporate as a storage form-factor option.

A hybrid energy system with photovoltaics, wind turbines, and fuel cells connected to an unstable grid was optimally designed in Egypt [22]. Hybrid Firefly and Harmony Search Optimisation (HSO) approaches were utilized to determine the optimal size of each component for the lowest net present cost after the subsystems and power management system model was proposed. The results were validated using the Particle Swarm Optimization approach. The findings show that the suggested approach is more practical than the grid. The tariff cost in Egypt is 0.1 \$/kWh, however, the LCOE is 0.0628 \$/kWh. Reference [23] demonstrated how to model, simulate, and optimize fuel cells and solar photovoltaic systems to meet the energy needs of 150 houses in the United Arab Emirates. In addition to meeting the daily and annual energy needs of the residential neighborhood, the proposed system should reduce the quantity of waste, increase the percentage of clean energy in the power mix, have a low LCOE, and have minimal environmental impact. The proposed off-grid hybrid renewable power system has an LCOE of 0.145 \$/kWh, is environmentally sustainable, generates 40.2% of its energy from renewable sources, and emits no carbon dioxide.

Most research in this field has focused on modeling, sizing, and integrating GES as a standalone energy storage solution, while little attention has been given to its hybridization with other storage systems such as hydrogen. None of the past studies have addressed the integration of hybrid energy storage incorporating gravity energy storage with hydrogen storage. The incorporation and optimal operation of a hybrid hydrogen/GES system into a solar/wind power plant to substitute a USC coal-fired power plant have not been investigated in previous studies. Indeed, existing studies either consider PV-WT-H₂ without GES, or PV-GES without H₂, or analyze GES without hybridization, often with hourly energy balance only and no strict reliability constraints. None co-optimize PV-WT-H₂-GES sizing and dispatch to replace a coal unit off-grid at LPSP = 0% while quantifying the separation of roles observed

in dispatch. This research addresses this gap by investigating the simultaneous optimization and energy management of a hybrid PV/WT/H₂ system combined with a storage solution to ensure a stable and efficient energy balance. We provide this integrated framework and evidence on a full meteorological year.

The main contributions of this paper are as follows:

- First time exploration of a hybrid GES and hydrogen storage system, marking a novel approach in energy storage integration.
- Development of an optimisation model to determine the power plant's capacity, tailored to satisfy a predefined load demand while leveraging key techno-economic performance indicators.
- Development of an innovative simultaneous optimisation and energy management strategy, custom-designed for a hybrid PV-wind-GES-hydrogen system, ensuring precise and efficient control of energy supply and demand.

Relative to prior PV-WT-H₂ or PV-GES studies, we present : (i) a joint sizing and dispatch optimization of PV-WT-H₂-GES under a 0% LPSP reliability target for an off-grid coal-replacement plant; (ii) a physics-aware GES model integrated into the EMS, which exposes the (GES) vs (H₂) roles from 8,760-hour dispatch; (iii) a Morocco case study that quantifies the LCOE at plant scale; and (iv) a single simultaneous optimization with control workflow, aligning techno-economics and reliability.

The paper is structured as follow. Section 2 reviews HRES design and storage technologies. Section 3 details the optimization and control frameworks. Subsequent sections present the case study, results, and conclusions.

2. System configuration

To replace the energy generation of a coal-fired power plant with a hybrid renewable energy system, this study examines a 624 MW power station currently operating with ultra-supercritical coal technology. The proposed solution integrates a hybrid energy storage system combining hydrogen, and gravity energy storage, and batteries to address the variability of renewable power generation. The study focuses on the simultaneous optimization and control of an off-grid hybrid PV-Wind-Hydrogen-GES system, proposed as a sustainable alternative to the ultra-supercritical coal-fired power plant in Morocco. It aims to determine the optimal sizing and capacity configuration of this hybrid storage system while ensuring its seamless integration into a renewable energy-based power plant. To achieve this goal, models for the power generation of solar and wind energy sources were developed, enabling a cost-effective and reliable energy system design. A schematic representation of the hybrid system, illustrating the power flow directions, is provided in Fig. 1.

The renewable energy sources considered in this study include solar and wind energy. An off-grid operation of the system is used. This means that the entire energy requirement must be met by the hybrid system renewable generation, supported by a hybrid energy storage system partitioned between hydrogen storage and GES system. Whereas solar PV power is routed through a DC-AC converter, wind turbines provide power straight into the distribution station through the AC grid. Given the intermittent nature of these renewable sources, loss of power supply is anticipated. To improve system dependability, the power plant incorporates a hybrid hydrogen-GES storage technology.

Specifically, when excess renewable energy is produced, an electrolyzer uses this energy to produce hydrogen through water electrolysis for later storage and use. Meanwhile, the GES system stores energy in potential form by lifting a mass to a higher elevation. Conversely, when renewable energy generation falls short of demand, the stored hydrogen is converted back to electricity via fuel cells, and the GES system releases energy through the controlled descent of the mass. The primary objective of this work is to determine the optimal hybrid storage system

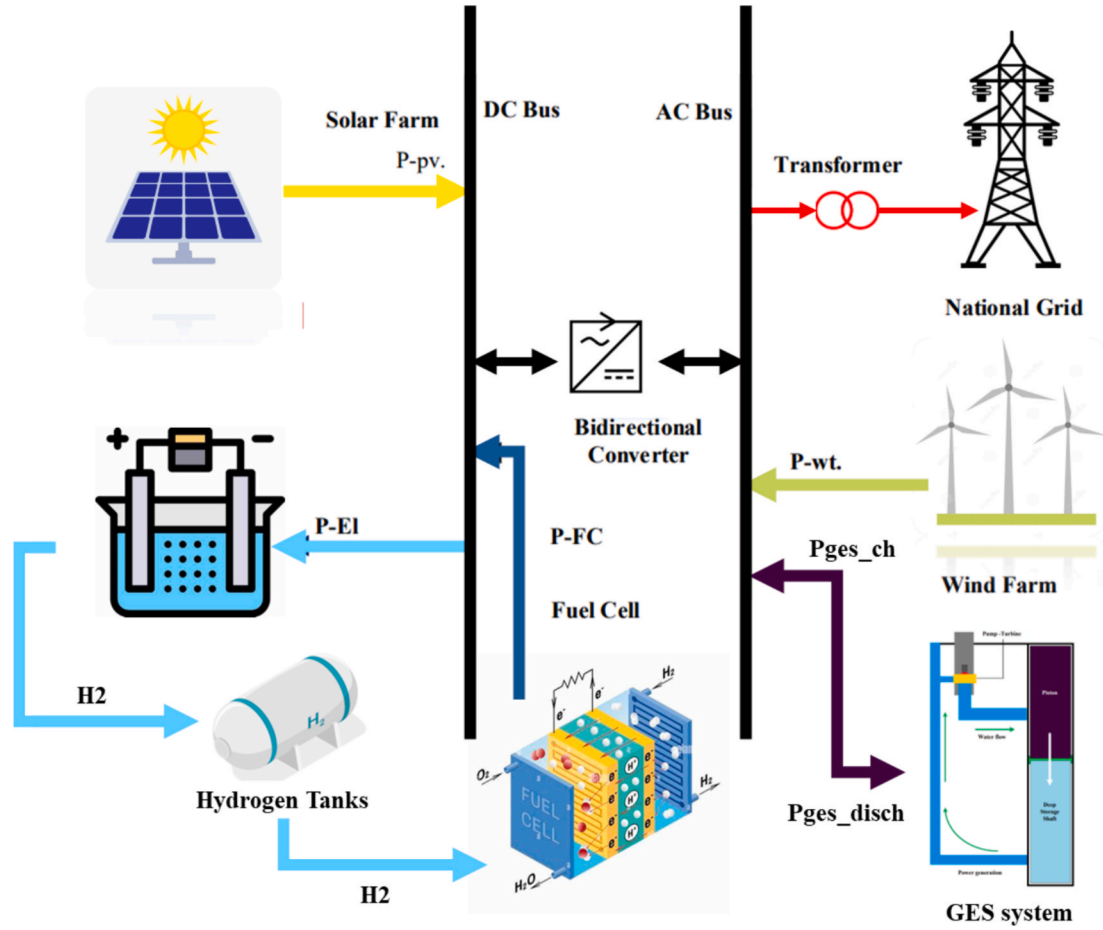


Fig. 1. Synoptic representation of the new PV/WT/H2 /GES hybrid energy system

capacity and renewable energy production design that meets load demand while adhering to reliability constraints.

3. Renewable hybrid system modeling

Modeling plays a crucial role in accurately evaluating the performance of the hybrid system. Below is a detailed representation of the key components within the hybrid energy chain.

3.1. Photovoltaic model

Solar photovoltaic technology harnesses sunlight and converts it into electricity through the photovoltaic effect. When photons strike a semiconductor material, energy is transferred to electrons, which creates an electric potential difference. The layered structure of solar panels, with distinct positive and negative regions, facilitates this process, and wires connected to these layers allow the generated electricity to be used [20].

In this research, we employed the SunPower Maxeon 6 (Model: SPR-MAX6-440-COM), widely recognized for its advanced design and high performance. Measuring 1762 mm by 1046 mm and 35 mm thick, each panel delivers a maximum power output of 440 W. This makes these modules among the most efficient choices available for renewable energy installations. The PV system production is dependent on solar irradiation and ambient temperature, as described by Eqs. (1), where, $T(t)$, $T_{STC} = 25^\circ\text{C}$, and $T_{air}(t)$, are cell, the reference, the ambient temperatures, respectively. The coefficient temperature is $k = -0.0029 / ^\circ\text{C}$. V is the wind speed, and $P_{PV}(t)$ is the PV power generation. E_p is the maximum output power under standard test conditions (STC), and $G_f(t)$,

$G_{STC} = 1000 \text{ W/m}^2$, are the actual and the rated solar radiation under STC, respectively. $H(t)$ is the Hourly Humidity [21].

$$\begin{cases} P_{PV}(t) = E_p \frac{G_f(t)}{G_{STC}} [1 + k(T(t) - T_{STC})] \\ T(t) = T_{air}(t) + 0.0318 \times H(t) \times (1 + 0.031 T_{air}(t)) \times (1 - 0.042 \times V) \end{cases} \quad (1)$$

The SunPower Maxeon 6 module leverages advanced Interdigitated Back Contact (IBC) cell technology, which eliminates front-side metal shading and enhances light absorption. Its superior temperature coefficient and low degradation rate ensure consistent performance even under varying environmental conditions (Table 1). This makes it an ideal choice for the proposed hybrid PV–Wind–Hydrogen–GES system.

3.2. Wind turbine model

Wind power essentially originates from solar energy. When solar

Table 1
Specifications of PV module

Parameter	Value
Cell Type	Monocrystalline IBC
Current at Max Power (A)	10.9
Voltage at Max Power (V)	40.5
Short Circuit Current (A)	11.0
Max Power (W)	440
Open Circuit Voltage (V)	48.5
Size (L × W × T) (mm)	1762 × 1046 × 35
Efficiency (%)	22.8

radiation is absorbed by the atmosphere, it causes variations in temperature and pressure that drive the movement of air masses and lead to the accumulation of kinetic energy. Wind speed and its fluctuations are the primary factors influencing wind power, which makes wind inherently intermittent and difficult to predict [22]. Air density also plays a significant role; for example, a temperature drop of 10 °C at a given wind speed can boost a turbine's power output by as much as 3%. The turbine is paired with a generator to produce alternating current in a stand-alone system that is supported by a backup energy storage mechanism, in this case the GES and FC.

In this study, the Vestas V164-9.5 MW, one of the most advanced and high-performance wind turbines on the market, is utilized (Table 2). This turbine is designed for both onshore and offshore applications and is renowned for its exceptional efficiency and reliability. It begins operation at a cut-in wind speed of 3 m/s and ceases operation at a cut-out speed of 25 m/s to ensure safety and longevity. Equation (2) describes the power generated by the wind system. When the wind speed (v) exceeds the cut-in speed (v_{cut-in}), the wind turbine's power generation (P_w) begins to operate. When the wind turbine reaches the nominal speed (v_{rate}), it attains its nominal power ($P_{wind,rate}$). When the wind speed hits the cut-out speed ($v_{cut-out}$), the wind turbine stops working for safety purposes. N_{wt} is the number of wind turbines [23,24].

$$P_w = \begin{cases} 0 & v < v_{cut-in} \\ \frac{N_{wt} \times P_{wind,rate} \times (v^3 - v_{cut-in}^3)}{v_{rate}^3 - v_{cut-in}^3} & v_{cut-in} < v < v_{rate} \\ N_{wt} \times P_{wind,rate} & v_{rate} \leq v \leq v_{cut-out} \\ 0 & v_{cut-out} \leq v \end{cases} \quad (2)$$

3.3. Gravity storage system generation and conversion model

Gravity energy storage system, schematized in Fig. 2, is a promising storage technique that has recently caught experts' attention because of its many benefits. The system consists of a heavy piston and a container connected to a pump turbine via a pipe. To store energy, excess electricity is used to convert electrical energy into gravitational potential energy by pumping water and raising the heavy piston within its container [25]. The system reverses the process in the energy release phase. The water is forced back into the pipe by the descending motion of the piston caused by gravity, where the turbine generator transforms the kinetic energy into electrical energy [21,26].

Equation (3) quantifies GES energy capacity; where E is the energy capacity in joules (J), g is the gravitational acceleration (m/s^2), h is the lifting height (m), mr is the pistons' relative mass (kg), and μ is storage efficiency [27,30].

GES capacity model is derived directly from piston–water hydrostatics. Diameter, height, and densities are explicit design variables. This physically parameterized approach is more realistic than using a fixed round-trip efficiency block, because it preserves geometry–capacity scaling. Parameters can be determined based on experimental validations of GES hydraulics [28,29].

Table 2
Wind Turbine Characteristics

Parameter	Value
Rated Power	9.5 MW
Rotor Diameter	164 meters
Rated Wind Speed	13 m/s
Swept Area	21,124 m ²
Cut-out Wind Speed	25 m/s
Cut-in Wind Speed	3 m/s
Efficiency	Up to 50% (theoretical maximum)
Generator Type	Permanent Magnet Synchronous
Operational Temperature	-20°C to 40°C
Hub Height	105–140 meters (adjustable)

$$E = \mu \times mr \times g \times h \quad (3)$$

This equation can be reformulated by considering the water and piston densities (ρ_e and ρ_p), together with the piston diameter (d) and height (H):

$$E = \mu \times (\rho_p - \rho_e) \left(\frac{1}{4} \pi d^2 H \right) \times g \times H \quad (4)$$

where H is the container height. The mechanical power stored (P_{ch}) and discharged (P_{disch}) from the system at any given moment are expressed as in Equation (5). They are dependent on the density difference of water and the heavy piston ($\rho_p - \rho_e$) and also the efficiency during the charging (η_{ch}) and the discharging phases (η_{gen}) of GES. Qch is the flow rate (m^3/s) and z is the height of water (m).

$$\begin{cases} P_{ch} = \frac{(\rho_p - \rho_e) \times g \times z \times Qch}{\eta_{ch}} \\ P_{disch} = \eta_{disch} \times (\rho_p - \rho_e) \times g \times z \times Qch \end{cases} \quad (5)$$

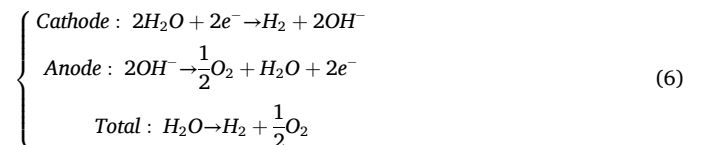
In comparison with conventional battery storage systems, GES systems offer several distinct advantages: they have a cycle life exceeding 50 years, maintain stable performance with minimal degradation in efficiency, and use environmentally friendly materials, such as recycled composites or waste. Advances in GES technology have demonstrated round-trip efficiencies of 80–85%, comparable to pumped hydro storage but without geographical limitations. Because of its long lifespan and minimal operating and maintenance costs, this system has a competitive leveled cost of storage (LCOS). The main advantage of GES is its flexibility and scalability; it can be installed everywhere and integrated into pre-existing infrastructure, including buildings and abandoned mines. Its ability to be built underground also contributes to its great social acceptance and minimal influence on the landscape. There is no risk of fire or explosion with GES.

Hybrid systems combining GES with hydrogen storage are both short- and long-term energy storage solutions. Although hydrogen storage offers excellent energy density for seasonal storage, its relatively slow response time (on the order of minutes to hours) makes it less suitable for rapid power fluctuations. In contrast, GES can adjust to power imbalances in a matter of seconds, but it has a lower energy density. The purpose of this study is to investigate the optimal synergy between these technologies, where GE systems handle high-frequency power fluctuations and hydrogen storage manages long-duration energy deficits. A hybridized power system is particularly crucial in off-grid renewable energy systems since they must maintain a consistent power supply despite being intermittent.

3.4. Hydrogen generation and conversion model

3.4.1. Electrolyzer

Through the decomposition of water when a direct current passes through it, water electrolysis generates hydrogen. A variety of water electrolysis technologies could be employed. One of the most developed and extensively utilized technologies is alkaline technology. This technique can generate hydrogen that is almost 99.989% pure. Due to the lower CapEx (1200 \$/kW vs 1000 \$/kW) and suitability for steady large-scale operation, alkaline is chosen over PEM. Equation (6) illustrates the use of alkaline electrolyzers to break down water [31].



Hydrogen is produced when there is excess renewable electricity gen-

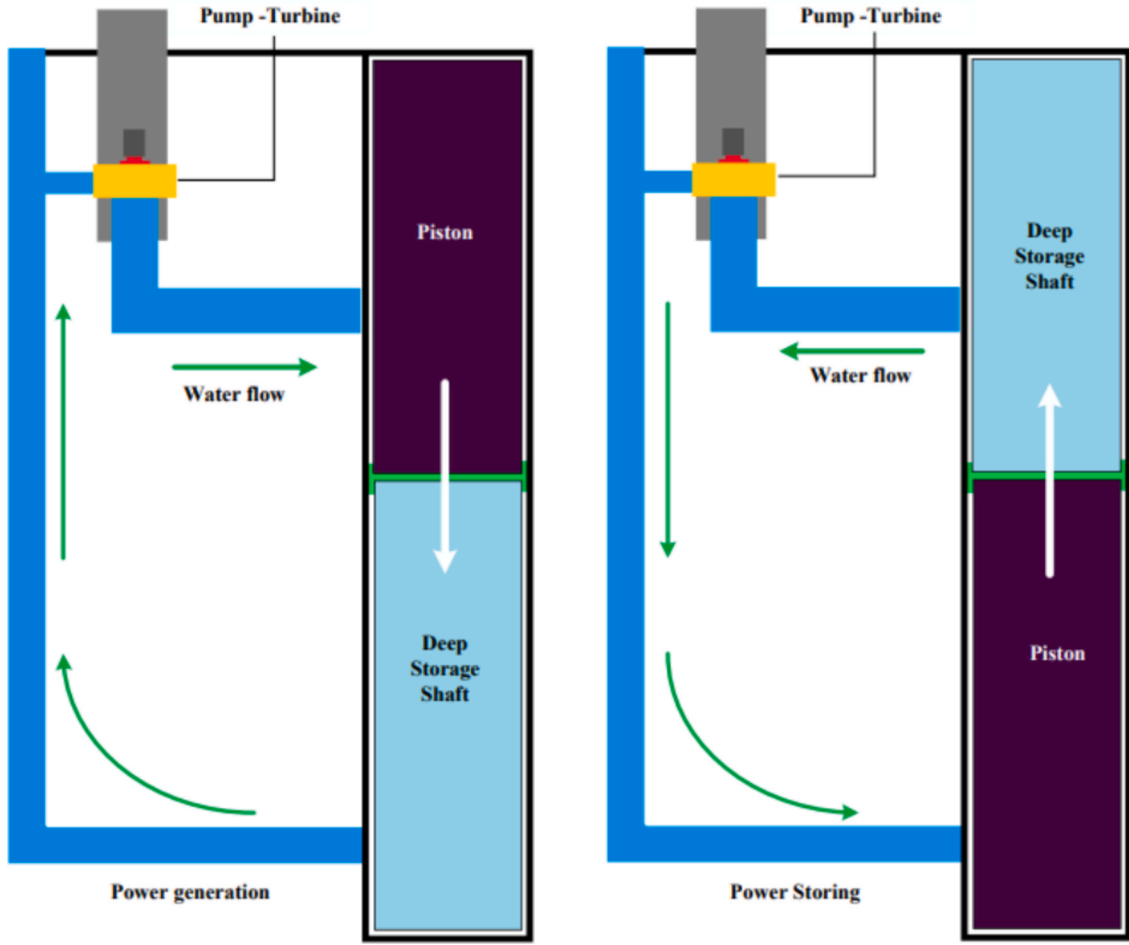


Fig. 2. Illustration of GES system

eration. The amount of hydrogen as a function of the available renewable energy can be determined considering the total amount of surplus energy and the energy that will be used to charge GES system as in equation (5). Efficiency $\eta_{elz} = 77\%$ (HHV basis) is consistent with current commercial systems.

$$P_{H_2}(t) = \eta_{elz} \times (P_b(t) - P_{ch}(t)) \quad (7)$$

Equation (8) calculates the actual amount of hydrogen produced $P_{H_2}(t)$ while accounting for the electrolyzer P_{Elz} rated power and the hydrogen tank's charge level.

$$P_{H_2}(t) = \min(E_{max-tank} - E_{tank}(t-1)) \min\{P_b(t) - P_{ch}(t) | P_{Elz}\} \times \eta_{elz} \quad (8)$$

Aqueous hydroxide solutions serve as the ion-conducting electrolyte in alkaline electrolysis, with potassium hydroxide being the most commonly utilized. The anode (which produces oxygen) and cathode (which produces hydrogen) electrodes are separated by a porous material that is filled with the aqueous electrolyte as the aqueous alkaline electrolyte is pumped through the cell. As the most advanced technology (highest manufacturing readiness level), this system is being manufactured at the highest volume with multiple 100MW+ installations in multiple decades.

These systems can have relatively low capital costs due to the low cost of the electrolyte and the use of lower-cost catalysts, separators, and porous transport layers in an alkaline environment [30]. Additionally, compared to other technologies, the comparatively high level of manufacturing readiness leads to economic advantages. Alkaline system development in the future may benefit from knowledge gained from similar industrial processes, such as chlor-alkali, which are already used

on an annual multi-GW scale.

Although alkaline electrolysis has been the largest-scale market, it has tended to focus on existing rather than emerging markets, where continuous operation has been the primary strategy. In the new global energy system, using an aqueous electrolyte has a number of drawbacks [30,31,32]. Using a corrosive (circulating) electrolyte has several drawbacks, such as the inability to function at differential pressures, which necessitates further compression of the hydrogen product; relatively high ohmic resistances because of the wide gaps between electrodes, which effectively limit operating current (power) densities; shunt currents between cells connected by a conductive electrolyte; and a limited capacity to react dynamically to variations in hydrogen production rates and start-up/shut-down cycling [33]. Additionally, these systems' electric to hydrogen (lower heating value) efficiency is usually restricted to less than 70% [34]. Its advantages include a low cost of materials, a long lifespan, and a proven manufacturing and supply chain.

Despite having faster response times and higher efficiency than alkaline electrolyzers, proton exchange membrane (PEM) electrolyzers are less suitable for large-scale deployments in this application due to their significantly higher capital costs (approximately \$1200/kW versus \$800/kW for alkaline) [35]. In commercial applications, alkaline electrolyzers have demonstrated operational lifetimes exceeding 80,000 hours, with predictable and cost-effective maintenance requirements [36]. Additionally, the relatively stable performance of the studied hybrid system reduces the need for fast load-following capabilities, which PEM systems excel at, making alkaline technology a more suitable option in this situation.

3.4.2. Hydrogen Storage Tank

The hydrogen energy that will be stored in the hydrogen tank $E_{\text{tank}}(t)$ is calculated using equation (9) and it is constrained to a minimum $E_{\text{min-tank}}$ and maximum $E_{\text{max-tank}}$, as stated in (10)

$$E_{\text{tank}}(t) = E_{\text{tank}}(t-1) + P_{H_2}(t) \quad (9)$$

$$E_{\text{min-tank}} \leq E_{\text{tank}}(t) \leq E_{\text{max-tank}} \quad (10)$$

The quantity of stored hydrogen $M_{\text{tank}}(t)$ in kg can be written as in Equation (11), where, HHV_{H_2} is the hydrogen gas higher heating value, which is usually taken as 39.7 kWh/kg.

$$M_{\text{tank}}(t) = \frac{E_{\text{tank}}(t)}{HHV_{H_2}} \quad (11)$$

3.4.3. Polymer exchange membrane fuel cell (PEMFC)

Fuel cells are electrochemical generators that produce power by reducing an oxidant at one electrode and oxidizing a reducing fuel at another. PEMFCs are among the most popular fuel cell technologies because of their extended lifespan, amazing efficiency, fast start-up time, and relatively low working temperature (<100 °C). Only heat, water, and electricity are produced by this fuel cell type [37]. The quantity of hydrogen drawn from the hydrogen tank and the fuel cell's efficiency (η_{FC}) determine how much energy the fuel cells can generate $P_{FC}(t)$ [38,39].

$$P_{FC}(t) = \eta_{FC} \times P_{\text{Tank-FC}}(t) \quad (12)$$

where, the flow of power from the hydrogen tank to the PEMFC is indicated by $P_{\text{Tank-FC}}$.

3.4.4. Hydrogen storage safety and practical considerations

Compression energy and storage pressures (200–350 bar typical for gaseous vessels) add an overhead to system efficiency. Storage options include modular high-pressure cylinders, underground lined caverns, or salt caverns where geology permits. Leakage and permeation, though low (<0.1% per day), require continuous monitoring and purge routines. Standard hydrogen safety measures include redundant leak detection, adequate ventilation, hazardous-area zoning, emergency vent stacks, and selection of embrittlement-resistant materials. These factors do not alter the dispatch optimization but define engineering boundaries. A full hazard and operability (HAZOP) assessment will be necessary at deployment

4. Energy management and optimization process

The core goal of the sizing algorithm is to provide a comprehensive framework for replacing coal-fired power plants with efficient, reliable, and economically viable hybrid renewable systems. To do so, the determination of the most effective configuration of the hybrid system, considering both energy generation (PV, wind, and FC) and storage (hydrogen tanks and GES system), is essential. A new techno-economic approach is employed, combining advanced energy management and optimization techniques. Indeed, the processes involved in achieving optimal system performance coordinate charge and discharge operations, set priorities for energy storage and generation systems, and balance energy supply and demand through a dynamic power management strategy (Fig. 3).

The proposed optimization approach focuses on both the technical and economic performance of the hybrid system by applying two key criteria, i.e., reliability criterion and economic criterion. The system's reliability is evaluated using metrics such as the Loss of Power Supply Probability (LPSP), which assesses the likelihood of insufficient power generation, and the Relative Excess Power Generated (REPG), which measures the system's capacity to handle surplus energy efficiently. The economic criterion, on the other hand, assesses the economic viability

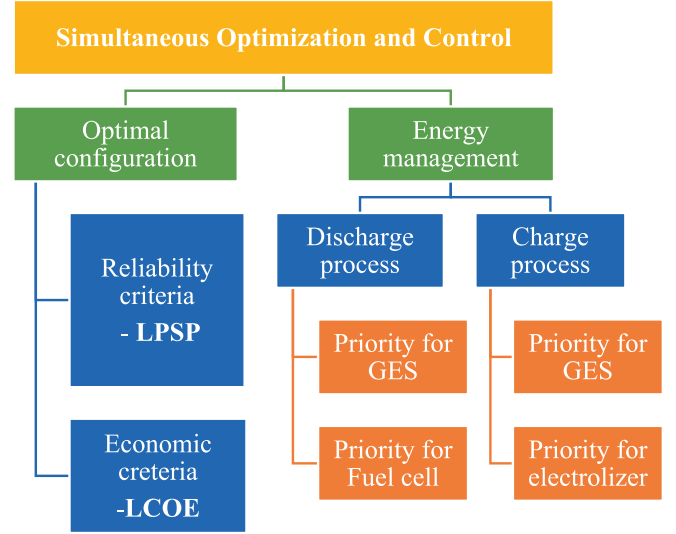


Fig. 3. Simultaneous Optimization and Control approach

through the levelized cost of electricity (LCOE).

The optimization framework simultaneously integrates optimization and energy management by prioritizing different energy storage systems, in our case, GES and hydrogen (Electrolyser (EL)/ fuel cell (FC)), based on real-time power management requirements in both deficiency and excess power modes.

The energy flow management in the hybrid system, illustrated in Fig. 4, operates as follows:

First, using Equations (1) and (2), the difference between the energy demand and the (PV) + (WT) output is determined.

- If $P_B(t)=0$, the hybrid energy sources match the demand exactly.
- If $P_B(t) > 0$, the combined output of the PV and wind systems exceeds the demand. In this case, the surplus energy is directed to charge the GES system, as described by Equations (11). Any remaining excess energy is then used to power the electrolyzer, which generates hydrogen, following Equations (3-6).
- If $P_B(t) < 0$, the energy demand surpasses the supply from the PV and wind systems. The deficit is compensated by the storage system, where the GES discharges first (Equation (11)) before the fuel cells step in (Equations (7-10)). If the hydrogen levels in the storage tank drop below the minimum threshold, the hybrid system can no longer meet the demand, and the Loss of Power Supply Probability (LPSP) is evaluated.

The Device-Level Primary Control acts on millisecond-to-second timescales and turns the EMS targets into actual device actions. Local controllers regulate power to the set-points: a hydraulic governor for the GES pump–turbine, a DC–AC converter for the fuel cell, and a DC power/current controller for the electrolyzer. Simple droop/PID control keeps bus voltage and frequency stable. Each interval, the EMS forms a power balance (PV + wind – load) and, based on that number, sends commands that are mutually exclusive for the GES (either charge or discharge), plus power set-points for the EL and FC while respecting state-of-charge and hydrogen inventory limits. The dispatch rule is straightforward: with surplus energy, charge the GES first and send any remaining power to the electrolyzer (curtail only if both are at limits); with deficit, discharge the GES first and then call on the fuel cell. A small deadband prevents rapid toggling, and a configurable reserve margin on GES/FC lets the plant absorb short fluctuations.

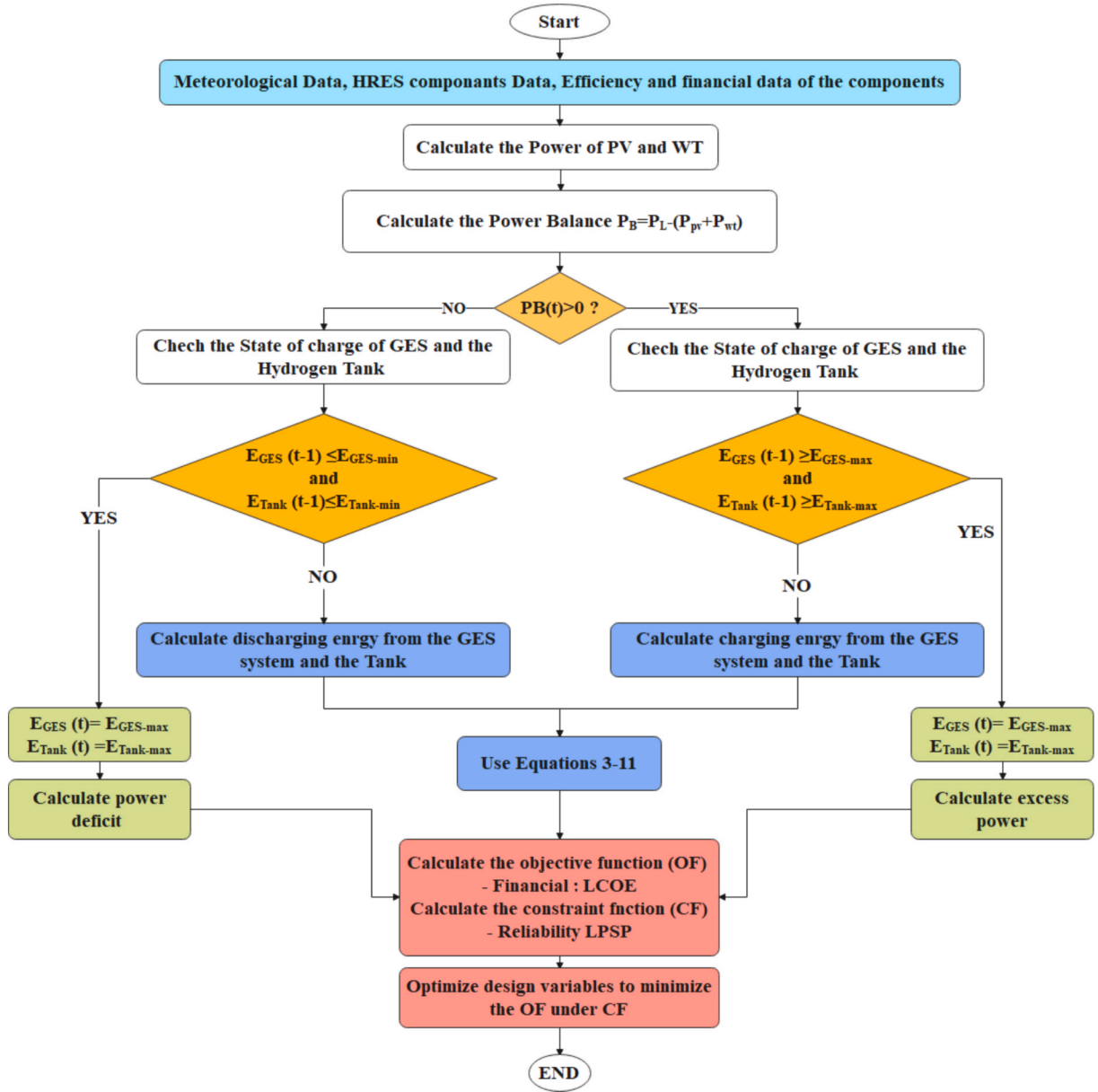


Fig. 4. energy management flowchart

4.1. Objective function

The hybrid PV/WT/GES/H₂ model's cost of energy (COE) is minimized in the developed model. The objective function can be found in Equation (15). The COE is computed using life-cycle cost analysis (LCCA). Equations (13)–(15) are used to calculate the project's total cost, evaluate the techno-economic performance of several configurations, and determine which is the most cost-effective. [40].

We take into account the capital costs associated with each component, such as PV, wind, EL, FC, and GES, when calculating the hybrid power plant's LCC. Operating and maintenance costs are also included in the LCC and are computed as a percentage of the original investment. The cost of component replacement, recycling, and disposal is also factored into the LCC. Details of the expenses related to each component are shown in Table 3. It is important to note that when evaluating the life cycle cost of GES, disposal, and recycling costs are not considered due to the lack of established demonstration plants [40].

$$C_{Cap-i} = Cap_{PV} + Cap_{GES} + Cap_{WT} + Cap_{ELz} + Cap_{FC} + Cap_{H2Tank} \quad (13)$$

$$C_{O\&M-i} = C_{O\&M-PV} + C_{O\&M-GES} + C_{O\&M-WT} + C_{O\&M-ELz} + C_{O\&M-FC} + C_{O\&M-H2Tank} \quad (14)$$

$$COE = \frac{\sum \frac{r(1+r)^{t_i}}{(1+r)^{t_i}-1} \times (C_{Cap-i} + C_{O\&M-i} + C_{REP-i} + C_{DISP-i})}{E_{Output}} \quad (15)$$

Where (C_{Cap-i}) is the capital expenditures (\$), Cap_{PV} , Cap_{GES} , Cap_{WT} , Cap_{ELz} , Cap_{FC} , and Cap_{H2Tank} are the capital cost of the PV system, gravity energy storage, wind turbines, Electrolysers, fuel cell, and hydrogen storage tank, respectively. $C_{O\&M-i}$ is the total operation and maintenance costs.

$C_{O\&M-PV}$, $C_{O\&M-GES}$, $C_{O\&M-WT}$, $C_{O\&M-ELz}$, $C_{O\&M-FC}$, and $C_{O\&M-H2Tank}$ are the operation and maintenance costs of the PV system, gravity energy storage, wind turbines, Electrolysers, fuel cell, and hydrogen storage tank, respectively. C_{REP-i} and C_{DISP-i} are the operation and maintenance costs replacement and disposal costs (\$).

Table 3
Financial assumptions [41,42]

Component	Parameter	Value
Electrolyzer	Capital cost	1000 \$/kW
	O&M cost	14.48 \$/kW
	Efficiency	77%
	Lifetime	20 years
Fuel Cell	Capital cost	600 \$/kW
	Replacement cost	600 \$/kW
	O&M cost	13.43 \$/kW
	Efficiency	60%
Hydrogen Tank	Life span	10 years
	Capital cost	376.4×1.15 \$/kg
	O&M cost	2%
	Salvage cost	193.23 \$/kg
Wind Turbines	Life span	25 years
	Capital cost	2850000 \$/WT
	O&M cost	3% CapEx
	Lifetime	25 years
Photovoltaic panels	Capital cost	765\$/Panel
	O&M cost	1% CapEx
	Salvage cost	440\$/kW
	Life span	25
Gravity Energy Storage	Capital cost	0.56\$/Wp
	O&M cost	1% CapEx
	Life span	25
	Project lifetime	25 years
Economic parameters	r	6%
	Project lifetime	25 years

4.2. Constraint function

The primary objective of the hybrid RE system is to secure a cost-effective and dependable energy supply. Therefore, a preset percentage of Loss of Power Supply Probability (LPSP) constrains the proposed target function. This is a crucial indicator that calculates the likelihood that the system will operate poorly if load requirements are not fully met. The LPSP is calculated in Equation (16). The predetermined LPSP requirement is considered an input parameter in the optimization problem. The solution-design variables acquired in each iteration of the iterative optimization process are fed into the smart energy management system, which calculates energy surplus and deficit and provides the relevant LPSP value for the present solution. To assess whether the answer is workable or if feasible or need to be refined, this calculated LPSP is compared to the predefined LPSP requirement. In an effort to attain the lowest feasible cost of energy (COE), the simulation keeps running until the dependability restriction is satisfied.

$$LPSP = \frac{\sum_{i=1}^{168} (P_{load}(t_i) - (P_{pv}(t_i) + P_w(t_i) + P_{FC}(t_i) + P_{GESdisch}(t_i)))}{\sum_{i=1}^{8760} P_{load}(t_i)} \quad (16)$$

where $P_{load}(t)$ is the load power (kW), $P_{pv}(t)$, $P_w(t)$, and $P_{FC}(t)$ are the hourly production power of the PV, wind turbine and Fuel cell systems, respectively. $P_{GESdisch}(t)$ is the GES hourly discharged power.

4.3. Optimization algorithm

The aforementioned formulation presents a nonlinear restricted optimization problem with two objectives: designing the hybrid PV-WT-H₂-GES to have the lowest energy cost and guarantee a 100% reliability level. The Fmincon nonlinear programming solver, a compelling tool in the MATLAB optimization toolbox, has been used to tackle the issue. This solver is a well-established gradient-based nonlinear solver capable of handling continuous decision variables with nonlinear equality/inequality constraints. MultiStart ensures robustness to local minima by sampling multiple initial guesses. Compared to heuristic/metaheuristic approaches, this choice gives faster convergence, analytical gradient availability, and repeatability on standard hardware.

Fmincon is designed to handle the following types of optimization problems:

$$\begin{cases} \min_x & function(x) \\ & constraint(x) \leq 0 \\ & constraint_{eq}(x) = 0 \\ & L_b \leq x \leq U_b \end{cases} \quad (17)$$

where x is the design variables vector. *constraint*

The formulated optimization problem involves seven design variables (PV capacity, wind capacity, electrolyzer, fuel cell, hydrogen tank, and GES diameter/height) and multiple nonlinear constraints. The Fmincon solver in MATLAB was used with an interior-point algorithm. The computational complexity per iteration is approximately $O(n^3)$, dominated by Hessian updates, with convergence typically achieved within 150–200 iterations. The total runtime averaged 2.5 hours for a full-year simulation (8760 time steps). This demonstrates that the optimization framework is computationally tractable. The implementation costs are modest, relying on widely available software (MATLAB Optimization Toolbox) and standard research-grade hardware. Consequently, the method is applicable not only for academic studies but also for practical energy planning scenarios at utility scale.

5. Case study

To demonstrate the practical feasibility and effectiveness of the proposed off-grid PV-Wind-Hydrogen-gravity energy storage system as a sustainable alternative to an ultra-supercritical coal-fired power plant, this section presents a detailed case study based in Morocco. The study focuses on Safi, a specific region which is characterized by high solar irradiation, strong wind potential, and increasing energy demands. The case study location has annual solar irradiation $\sim 1,800$ – $2,000$ kWh/m² and coastal wind speeds 6–9 m/s, with seasonal complementarity evident in Fig. 6. Localized variability, such as summer calms or winter irradiation dips, is captured by the 8,760 hour dataset. Socio-economically, deployment leverages local wind/PV industries and emerging electrolyzer value chains, while GES shafts could repurpose brownfield or mining sites, minimizing land conflict. Water demand for electrolysis can be met by desalination with brine management. The hybrid's ability to provide firm renewable electricity supports Morocco's decarbonization, reduces fuel imports, and creates jobs in O&M and construction.

The optimal design, operation, and economic viability of the hybrid system are evaluated by analyzing real-world energy consumption patterns. Fig. 5 represents the load demand profile that should be met by the ultra-supercritical coal-fired power plant. The plant's load demand exhibits repeated peaks near its upper capacity limit (around 650 MW) and troughs that are substantially lower; dips can fall to roughly 450 MW at the lowest points. The mean load appears to hover around 550–580 MW. This suggests that the unit is routinely cycling between high-load and partial-load operation rather than maintaining a steady baseload. Indeed, it undergoes repeated cycles of ramp-up (to around 600–650 MW) and ramp-down (to around 450–500 MW). This cycling suggests the plant is being actively dispatched to follow demand fluctuations on the grid rather than simply running at maximum capacity. The variability (the difference between peaks and troughs) is on the order of 150–200 MW within relatively short intervals. This indicates significant operational flexibility, but it also implies higher mechanical and thermal stresses on the plant components compared to steady-state baseload operation.

Meteorological data of the study location are essential for the optimal design and energy management of the hybrid renewable power plant. Fig. 6 depicts the essential information about temperature, solar irradiation, and wind speed. The meteorological data were sourced from the European Commission's PVGIS (Photovoltaic Geographical Information System), which combines satellite and reanalysis data validated against ground stations. We extracted hourly values for a full year (8,760 h). Validation confirmed expected climatology: annual solar irradiation

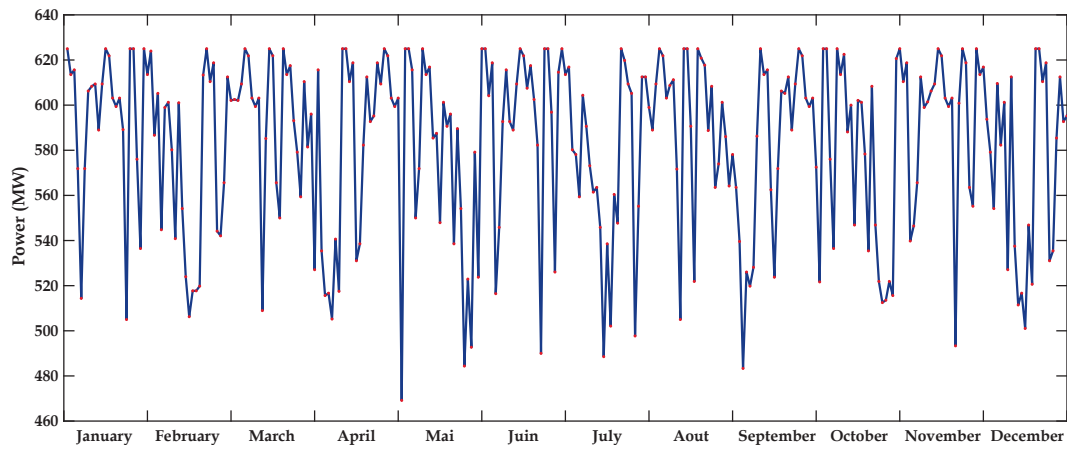


Fig. 5. Averaged hourly load demand covered by the Power plant over the year

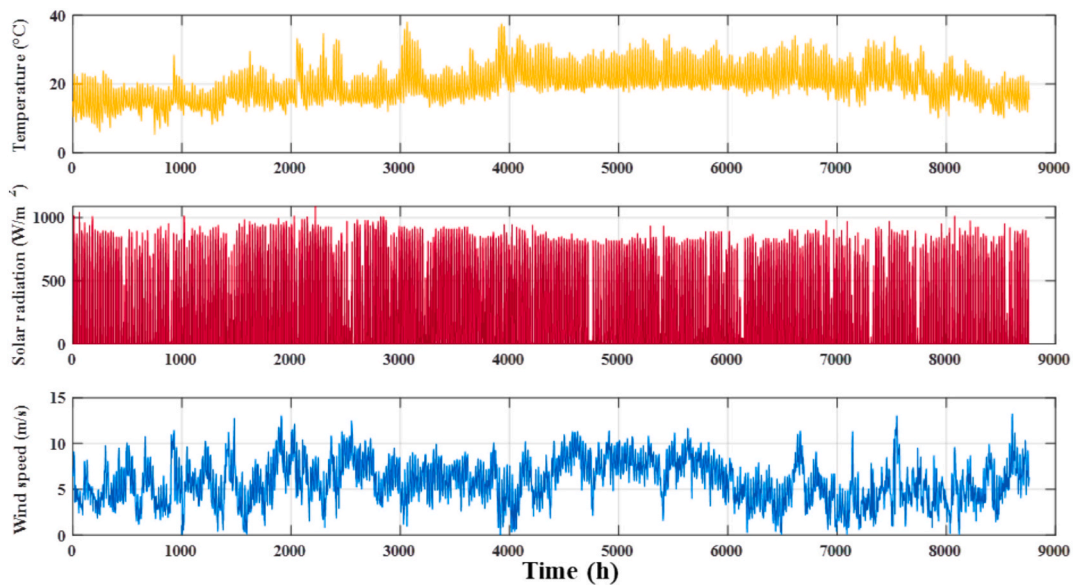


Fig. 6. Climatic data

$\approx 1,850\text{--}1,950$ kWh/m² and average wind speeds $\approx 6\text{--}8$ m/s, consistent with reported Moroccan resources. These steps ensure dataset robustness for system optimization and dispatch analysis.

Temperature data are primarily fluctuating between about 15°C and 38°C. The signal oscillates in a quasi-cyclical manner which suggests a daily and seasonal pattern. The solar irradiation (W/m²) extends roughly from 0 to 1200 W/m². The data exhibit repeated sharp peaks, occasionally surpassing 1000 W/m², followed by near-zero values, which is typical of daily solar cycles. Peak intensities often exceed 800 W/m², while overnight or during heavy cloud cover it drops to zero. Typical speeds spanning roughly 2 m/s up to around 15 m/s are observed. The wind speed varies more randomly but still shows moderate cyclical tendencies.

It can be observed that the substantial peaks in solar irradiation highlight the potential for photovoltaic power generation. The variability in wind speed also suggests a moderate to high wind energy potential, though intermittency must be managed and complementary resources are required (e.g., energy storage) to maintain grid reliability. The fluctuating profiles further indicate the need for flexible operation and possibly additional reserves.

6. Results and discussion

6.1. Optimization results

The optimization process was used to identify each source's ideal capacity in order to reduce the hybrid system's overall cost while meeting load demand requirements with a predetermined reliability level.

Fig. 7 illustrates the advancement of the optimization process aimed at achieving the ideal values of the design variables, hence minimizing costs while fulfilling 100% of the demand. The number of PV modules and wind turbines, as well as the GES system capacity, the hydrogen tank's capacity, as well as the fuel cell end electrolyzer power, are the optimal design factors determined by the developed model.

The algorithm's rising design variable values and resulting higher cost cause the objective function (COE) to exhibit an increasing trend. The system achieves the best possible value for the objective function after 180 iterations. For 100% reliability, the lowest cost achieved is 0.23€/kWh.

Table 4 shows the results obtained for the optimization problem with LPSP=0%. To meet the total load demand, the required capacities are 0.45 MW of solar energy, 1.88 GW of wind energy, and 37.9 MW of GES

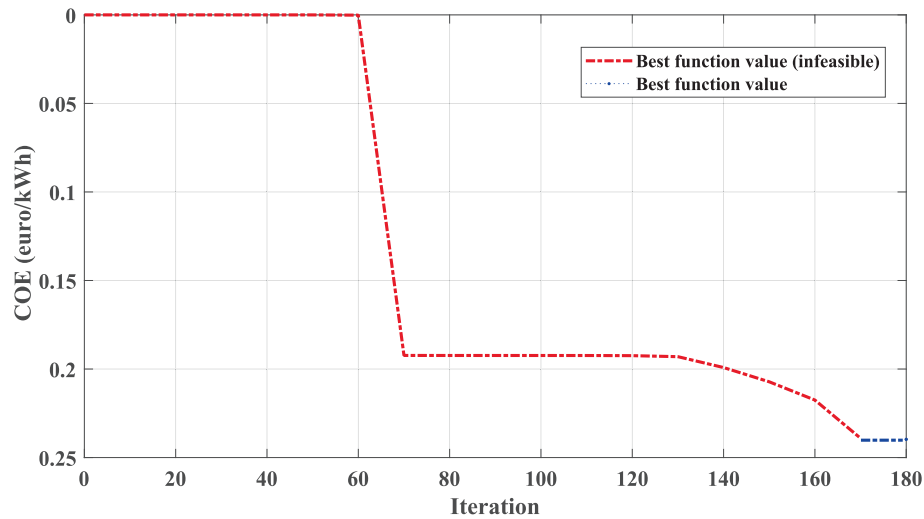


Fig. 7. convergence curve for the Fmincon optimization algorithm SWEAT

Table 4
Optimization algorithm results.

Variable	Value	Unit
Minimum LCOE	0.239	Euro/kWh
Number of PV modules	1005.268	Unit
Number of Wind turbines	594.093	Unit
GES H	714.964	m
GES D	5.190	m
P_{EL}	790	MW
P_{FCn}	650	MW
M_{tank}	2.60	tonne

storage, yielding a minimum LCOE of €0.23/kWh. The LPSP (%) represents the fraction of time the load is not fully met. Therefore, a higher LPSP relaxes reliability constraints. The minimum LCOE (€/kWh) is expected to decrease as system components are sized smaller. The number of PV modules and wind turbines tends to drop with higher LPSP, since the system does not need to supply 100% of demand at all times. For gravity energy storage, the height and diameter sizing may decrease because lower reliability requirements reduce total storage needed. The rated capacity of the electrolyzer is also lowered as less

hydrogen production is required for higher LPSP values. The nominal capacity of the fuel cell may also be reduced under lower LPSP. Hydrogen tank capacity can be smaller when total coverage is less strict.

The dynamics of energy supply and demand are shown in Fig. 8. The energy supply ranges from zero up to about 8 GW while the load demand is around 0.6–1.0 GW. The aggregated generation from PV, wind, GES, and fuel cells succeeds in covering the full load demand. Indeed, the power supply (blue) remains above the load (red) for all the plotted intervals. In a few instances, the supply curve approaches the load level; however, it still meets or slightly exceeds the required demand, which implies no load curtailment in those intervals (i.e., Loss of Power Supply Probability is effectively zero).

The large spikes in supply are likely driven by strong wind availability or mid-day peaks in solar output. The surplus energy can be transferred into storage (charging the GES or producing hydrogen) or curtailed if storage is full and there is insufficient export capacity. On the other hand, periods when the supply drops are due to reduced renewable input, but other sources, namely GES and fuel cells, compensate to maintain load coverage.

Fig. 9 depicts the optimized PV output. The curve repeatedly rises from near zero to peak values of around 4–5.5 MW. The power output shows a strong correlation with daytime hours (i.e., when the sun is most

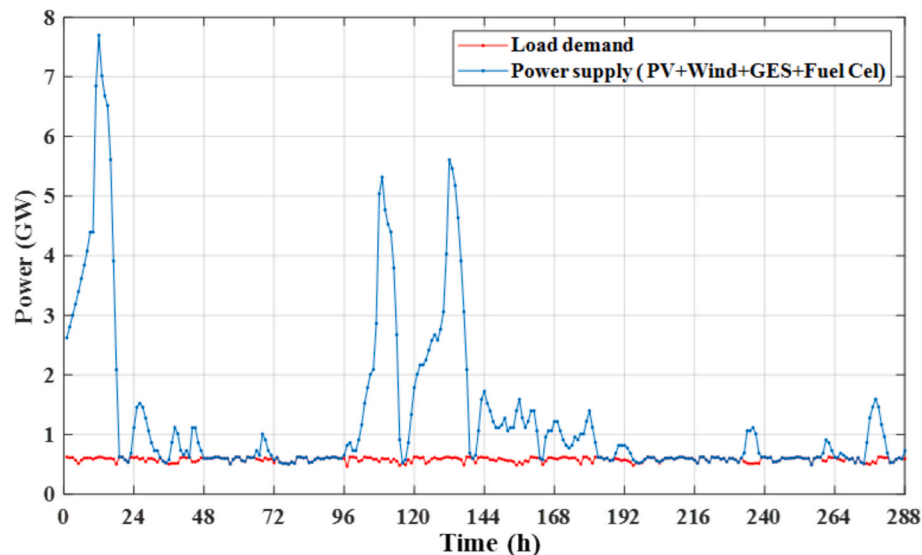


Fig. 8. Dynamics of energy supply and demand

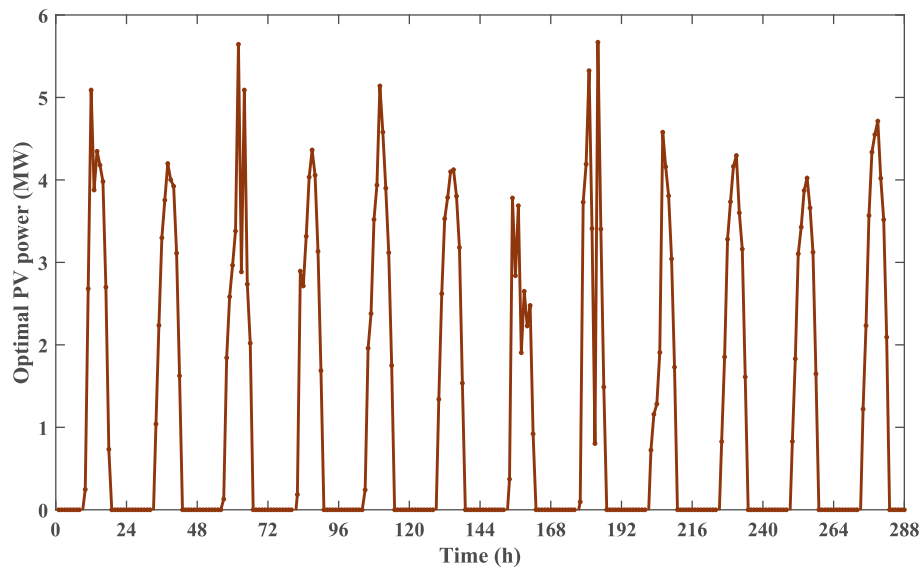


Fig. 9. Optimal PV output

intense). The consistent peak near 4–5 MW highlights the size of the PV array that the optimizer deems cost-effective for meeting the target objective (e.g., minimizing LCOE, ensuring 100% reliability). The importance of storage or complementary resources to manage PV fluctuations effectively is clearly demonstrated.

Fig. 10 illustrates the optimized wind power output over time. The values vary significantly from near zero to peaks approaching 7–8 GW. In the first approximately 10 hours, the wind output spikes to 7–8 GW, the highest level in the entire time frame. It then declines to around 3 GW before falling further. Another major spike occurs near the 100–120 h mark, reaching about 6–7 GW. Between 120 h and 300 h, wind generation oscillates mostly between zero and 2–3 GW. This may be due to moderate wind conditions. The presence of very high peaks suggests that storage or other dispatchable resources should be used during these surplus times.

The total power requirement over time in comparison with the net combined power from photovoltaic and wind generation together with the fuel cell and the gravity energy storage system output is shown in Fig. 11.

The PV + Wind curve has frequent spikes reflecting the intermittency of solar and wind resources. GES Exhibits bursts of output that complement renewables spikes coincide with times when the blue curve (PV + Wind) has dropped. This means that the storage system is discharging to meet demand. The FC offers moderate power contributions. The peaks are around a few hundred megawatts to approximately 600 MW at certain times. Its operation fills in gaps whenever PV + Wind output and GES output are insufficient to cover the load. At most time steps, the combined generation (PV + Wind + Fuel Cell + GES) is sufficient to meet the load. Indeed, during high renewable availability, any excess energy is stored by elevating the GES's piston or by producing and storing hydrogen. When solar and wind outputs drop, the GES or fuel cell ramps up to maintain uninterrupted power. The figure shows how multiple storage technologies can be orchestrated to ensure a nearly constant supply. Indeed, the presence of two storage pathways provides the flexibility needed to buffer against abrupt dips in PV + Wind.

Recurrent peaks in the PV + Wind curve well above the demand line indicate that the system is appropriately sized to handle the worst-case scenario of minimal renewable output at other times.

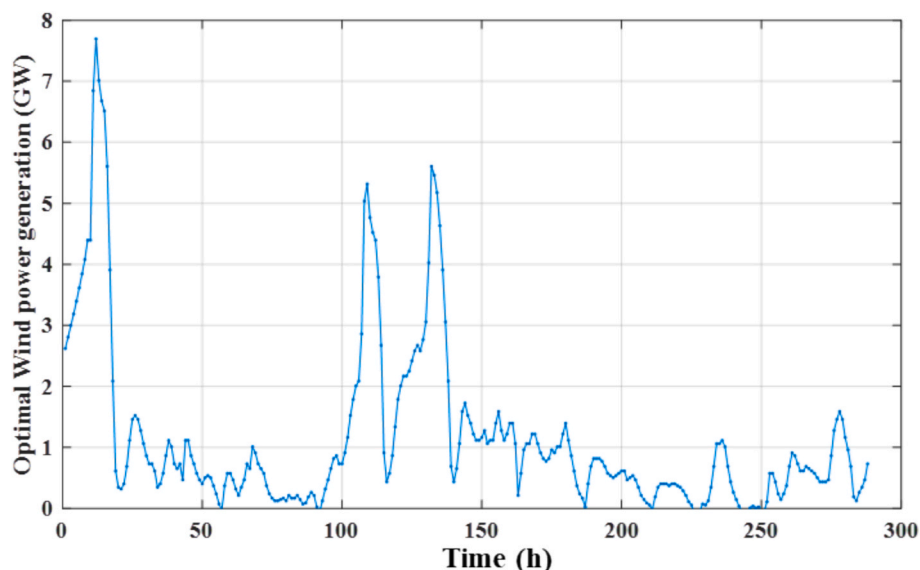


Fig. 10. Optimized WT generation

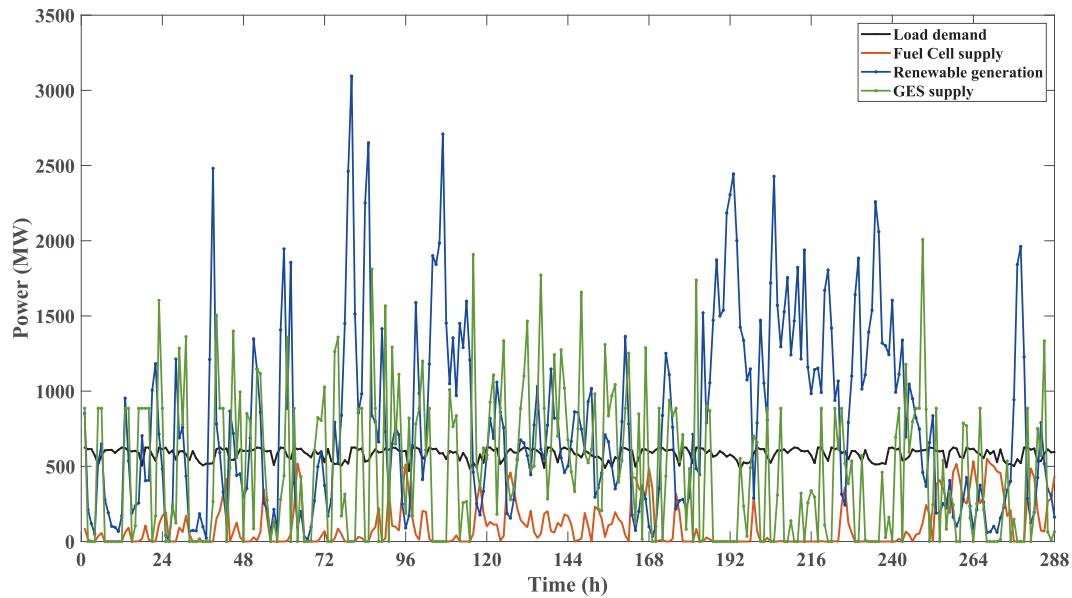


Fig. 11. Dynamics of energy supply and demand

The dynamic interaction between hydrogen production and renewable energy generation is shown in Fig. 12. Hydrogen production appears to follow renewable generation. Indeed, when there is abundant renewable power input, this power is directed toward hydrogen generation. In this case, H_2 production increases steadily to about 1 tonne. The stock of hydrogen exhibits a dip back down toward 0.7–0.8 t in periods of reduced availability of surplus renewables. Then it rises sharply to over 2.5–2.7 tonnes. The consumption of hydrogen is not significant because we start discharging the GES system. Indeed, it maintains a plateau around 2.5–3 tonnes for a period before gradually declining to approximately 1.5 tonnes. With GES near its upper bound, the EMS diverts the remaining surplus to the electrolyzer, so the H_2 tank mass continues to increase despite the downward trend in renewable generation. This behavior is consistent with the dispatch priority schematized in figure 4.

The fluctuations in hydrogen output likely mirror the variability of the energy sources that feed the electrolyzer. Higher production intervals occur during strong wind/solar conditions and full charge of

GES, whereas lower production or dips coincide with less renewable generation and the possibility of storing energy in GES.

Fig. 13 presents the GES storage level at each time step. The SoC varies between nearly fully charged (80–100%) and fully discharged (close to 0%). GES often maintains an SoC around 80–95%. Periodically, it decreases to almost 0% which corresponds to high demand periods where it discharges fully to meet the load or optimize costs. After a discharge event, the SoC returns to approximately 80–90% within a relatively short timeframe. This means that surplus renewable power (wind/solar) becomes available to refill the system.

The frequent transitions to nearly zero SoC indicate that GES is playing a critical balancing role and highlights the importance of having optimized and sufficient capacity to cover demand when primary renewable sources are insufficient.

Fig. 14 reflects fuel cell output over time, with power levels oscillating between nearly zero at some intervals and peaks reaching 600–650 MW at others. The fuel cell acts as a firming resource, ensuring a continuous electricity supply. Its output drops to zero during periods

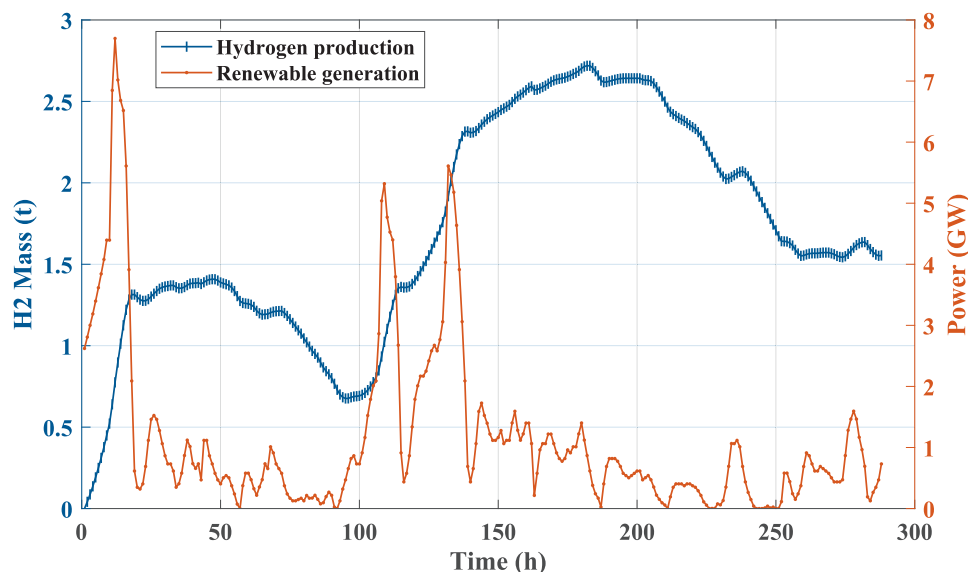


Fig. 12. Hydrogen Production Compared to Renewable Energy Generation

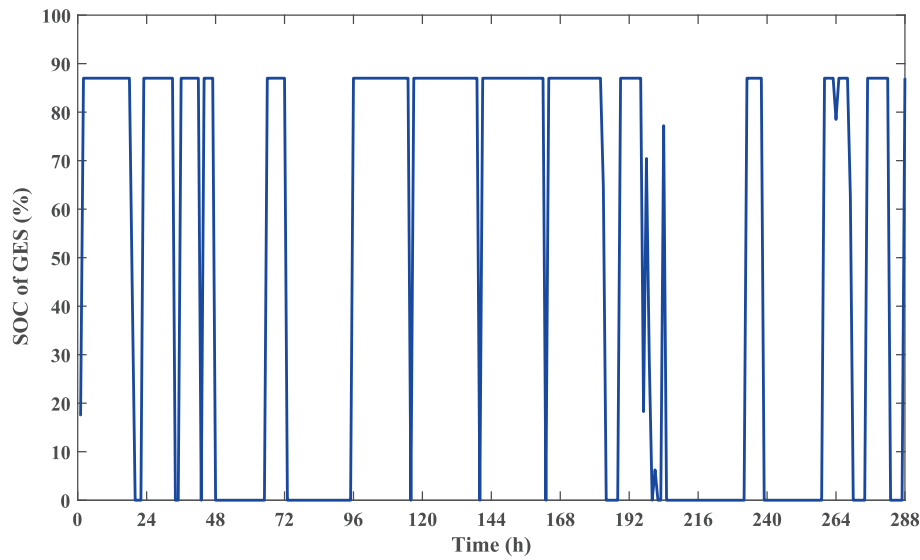


Fig. 13. GES state of charge

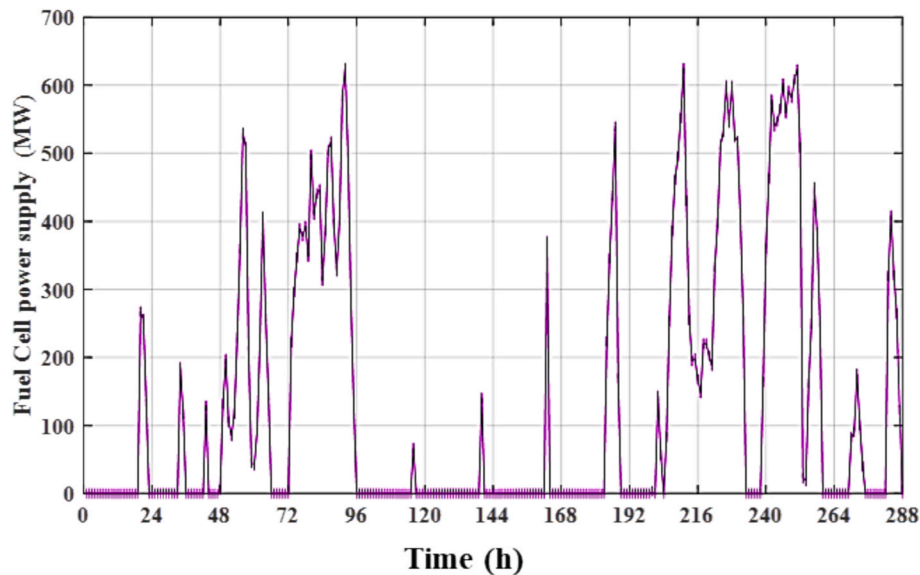


Fig. 14. FC power output.

when wind, PV, or storage sufficiently meet demand, making it more cost-optimal to keep the fuel cell idle. Clusters of high FC activity occur between 24 h and 96 h and after 190 h, with repeated spikes to 500–650 MW. The fuel cell's power is supplied by stored hydrogen, which is produced during periods of excess renewable energy generation.

The hybridization of hydrogen storage with gravity energy storage creates a complementary hybrid system that leverages the distinct advantages of each technology to address the limitations of standalone energy storage solutions. Since hydrogen storage has a high energy density, it is ideal for bridging long-term gaps in renewable energy generation due to its long-term energy retention capabilities (from days to seasons). While it can handle rapid power fluctuations, its relatively slow response time (minutes to hours for electrolyzers/fuel cells) and round-trip efficiency losses limit its flexibility. On the other hand, GES can provide ultra-fast response (seconds to milliseconds) and high cycle efficiency (80–90%), but it is constrained by its high energy density.

A clear operational synergy is demonstrated by the hybrid hydrogen-GES system. Due to its fast ramp-up time, GES responds first to high-frequency power imbalances during periods of excess renewable

generation, while surplus energy is diverted to hydrogen production. On the other hand, GES handles transient demand spikes during prolonged low-generation periods. This coordination reduces electrolyzer wear and minimizes energy curtailment, improving overall system efficiency compared with single-storage configurations.

GES and hydrogen storage work synergistically during periods of variable renewable energy. Fig. 8–14 shows the system transitioning through multiple operational modes over 72 hours. When solar generation exceeds demand during hours 12–18, the GES system responds immediately by storing the surplus, reaching full capacity. With the remaining excess energy, the hydrogen production system ramps up more gradually. This staged response prevents energy curtailment while maximizing storage capacity.

The system, on the other hand, draws energy from GES during the extended low-generation period from hours 48–60 to meet immediate demand spikes, while the hydrogen fuel cells provide steady baseload power during this period. In comparison with a hydrogen-only system, this coordinated response reduces cycling stress on the fuel cells, thereby extending their operational lifetime. Finally, GES reduces electrolyzer

wear and extends the lifespan of hydrogen components by absorbing large amounts of charge/discharge cycles. Therefore, hybridization results in extended component lifetimes.

In comparison with conventional coal-fired power plants, the hybrid system offers significant advantages. The optimized hybrid system achieves equivalent reliability at a LCOE of (\$0.23/kWh) with negligible fuel costs, compared to a typical coal plant with an LCOE of \$0.18–0.34/kWh [43]. Even more striking is the system's environmental benefit: whereas coal plants emit approximately 876 kg CO₂ per MWh, our system emits a negligible amount per MWh (comprising manufacturing and installation) [44]. The hybrid system is cost-competitive even without considering carbon pricing, based on our economic analysis. CO₂ certificate prices in Germany, for example, are expected to increase to between 175 and 375 €/t by 2045.

Hybrid energy systems also provide greater operational flexibility than conventional alternatives. The hybrid system can respond to load changes within seconds (through GES) and maintain efficiency across a 20–100% operating range while coal plants need 6–8 hours for cold starts and operate most efficiently at 80–90% of rated capacity. This system is especially useful in grid with high renewable penetration, as conventional baseload plants have difficulty ramping up and down rapidly.

6.2. Sensitivity analysis

To assess the robustness of the design and reveal cost–reliability trade-offs, we performed a one-at-a-time ($\pm 5\%$) sensitivity around the optimized baseline (LCOE ≈ 0.239 €/kWh).

Fig. 15 depicts the local sensitivity of the LCOE to $\pm 5\%$ perturbations in seven design variables at the optimized baseline. Bars show the absolute LCOE obtained when each variable is changed by -5% (blue) or $+5\%$ (orange) while all other variables are held at their baseline values. Variables are sorted from top to bottom by their “swing” (the larger of the two absolute deviations from the baseline LCOE). The vertical line marks the baseline LCOE. Text labels at the bar tips report the corresponding reliability outcome (LPSP), making explicit the cost–reliability trade-off.

The ordering of variables highlights which components most influence the LCOE in the neighborhood of the baseline. Capital-intensive supply and storage elements typically dominate the response (e.g., wind turbines, GES geometry), because their annualized costs scale directly with installed quantity or effective capacity. Increases of $+5\%$ in

generation or storage generally push LPSP toward zero but at the expense of higher LCOE, whereas -5% changes yield the opposite: slightly lower LCOE but a higher LPSP. This reveals the immediate cost–reliability trade-off. Electrolyzer power governs how quickly surplus can be turned into useful, dispatchable energy and how reliably deficits can be met later, so its $\pm 5\%$ perturbation moves both cost and reliability more than comparable changes in storage capacity or PV count. For the GES geometry, cost and usable capacity scale nonlinearly with height/diameter; therefore, even modest geometric changes can induce comparatively larger LCOE swings, which the tornado layout makes apparent. The hydrogen inventory rarely hits its upper or lower bounds over the year, so $\pm 5\%$ volume changes leave dispatch nearly unchanged. Same for PVs, in the $\pm 5\%$ neighborhood the system is conversion-limited, not PV-limited. It is worth mentioning that the strict 0% LPSP case illustrates feasibility for coal replacement, but practical planners may prefer slight reliability relaxation for cost savings.

Fig. 16 shows that LCOE is dominated by capital intensity and financing. The responses are near-linear and mostly symmetric. A $\pm 5\%$ change in wind CapEx shifts LCOE the most ($\approx -2.7\%/+2.8\%$), reflecting the large share of total investment tied up in turbines. The discount rate is next ($\approx -2.2\%/+2.1\%$), since a higher r inflates the present value of all long-lived costs. A second tier of drivers includes electrolyzer CapEx ($\approx \pm 1.0\%$), wind O&M fraction ($\approx +1.1\%$ for $+5\%$), and fuel-cell CapEx/replacements ($\approx +0.6\text{--}0.8\%$), consistent with their sizable but smaller cost shares. Items linked to PV (CapEx, O&M, salvage) and H₂-tank salvage barely move LCOE (near 0%), indicating these terms are marginal in the current scale mix.

6.3. Comparative analysis

To benchmark our results, we compared them with several recent contributions on hybrid renewable–storage systems. Xu et al. [45] optimized a PV–WT–H₂ system without gravity energy storage, achieving an LCOE of 0.091 \$/kWh when 5% LPSP can be acceptable, whereas the inclusion of GES increases costs to 0.23 €/kWh while ensuring stricter reliability (LPSP = 0%). Rehman et al. [12] analyzed an offshore PV–WT hybrid system for electricity and hydrogen co-production, demonstrating strong operational flexibility but without short-duration storage to mitigate rapid fluctuations. Amoussou et al. [29] investigated a PV–WT–H₂–battery configuration for replacing a heavy fuel oil plant, reporting a lower LCOE (0.145 \$/kWh) but at reduced reliability

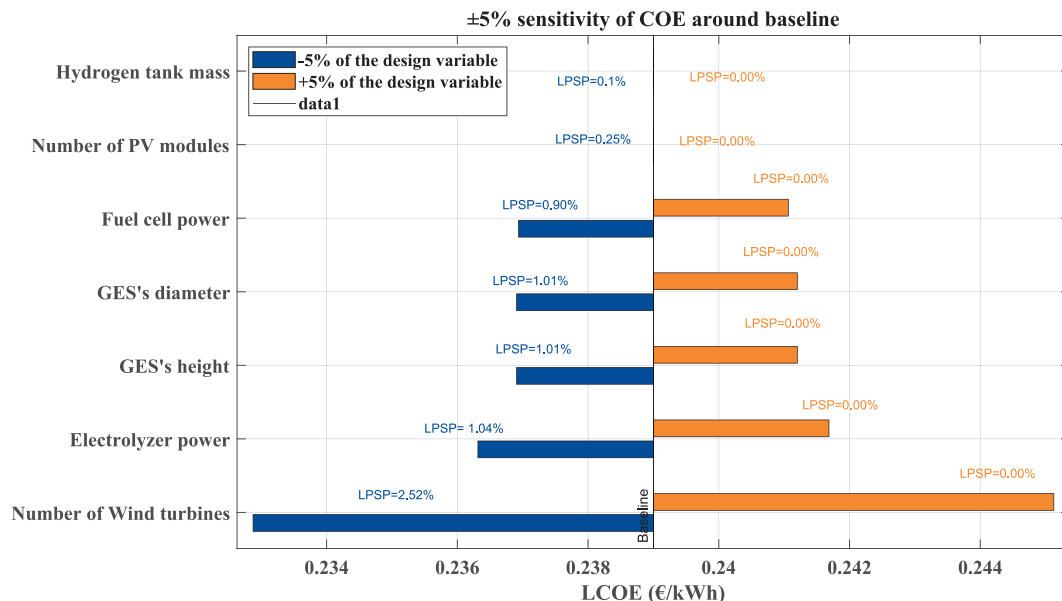


Fig. 15. One-at-a-time ($\pm 5\%$) sensitivity around the optimized design

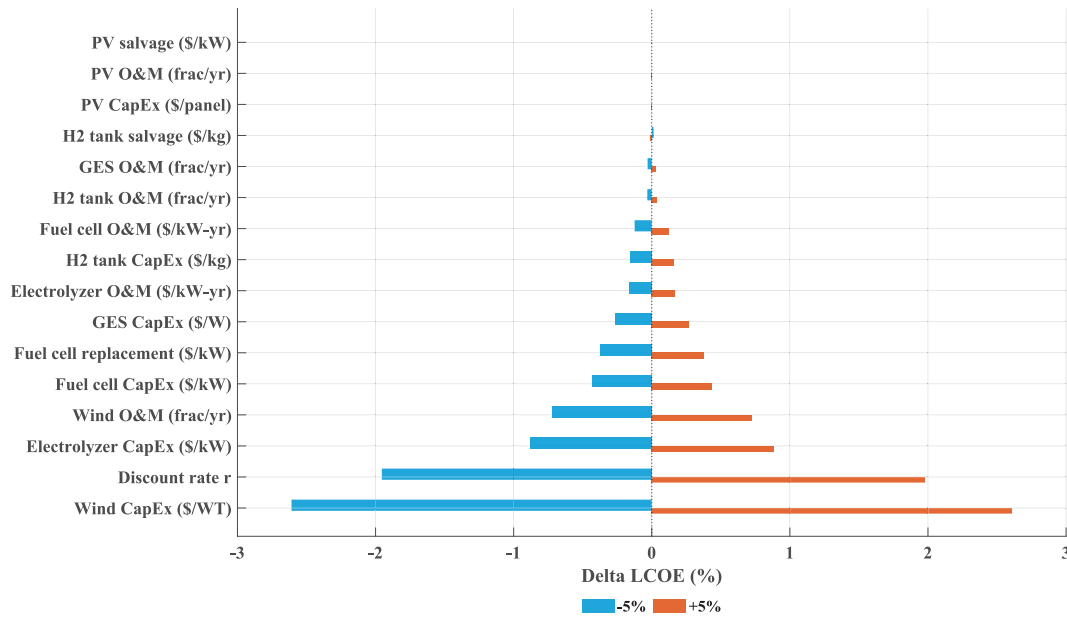


Fig. 16. One-at-a-time $\pm 5\%$ sensitivity on techno-economic inputs

($\sim 95\%$). Similarly, Emrani et al. [21,46] studied the PV–WT–GES system configuration, highlighting their fast-response capability but omitting long-term hydrogen storage. Moreover, Literature shows Li-ion excels at short, daily shifting, while H₂ provides economical energy capacity at multi-day/seasonal horizons—but with lower RTE. Recent gravity-storage work demonstrates low-degradation, high-cycle capability for intra-day balancing. Our work uniquely co-optimizes both hydrogen and GES technologies within a unified framework, demonstrating that their complementary roles (long-term vs. short-term balancing) allow the system to achieve 100% reliability while maintaining competitive costs. This dual-storage synergy, absent in previous studies, represents a key advancement of our proposed model. Table 5 contrasts our PV–WT–H₂–GES against (A) PV–WT–H₂ (no GES), (B) PV–WT–GES (no

H₂), and (C) PV–WT–Li-ion across LCOE, curtailment, storage duration, round-trip efficiency, degradation/cycling, and scalability.

6.4. Feasibility, limitations, and scalability potential

The optimized GES geometry (diameter ≈ 5 m; height ≈ 714 m) is within the constructability envelope of modern shaft-sinking/raise-boring techniques. Realization can follow two pathways: a single-shaft layout matching the optimized dimensions [26,46]; or a multi-shaft layout (e.g., 2–6 parallel shafts with smaller diameters) to ease civil works, enable staged commissioning, and provide redundancy [19]. Mechanical limits such as piston mass/velocity, seal wear, and penstock head loss are already captured at the dispatch level via rating

Table 5
Comparative benchmarking of storage topologies for PV–WT hybrids

Criterion	Proposed: PV–WT–H ₂ –GES	PV–WT–H ₂ (no GES)	PV–WT–GES (no H ₂)	PV–WT–Li-ion
Primary strengths	Combines seasonal/long-duration (H ₂) with fast, high-cycle gravity for firm supply and reduced curtailment/cycling	Excellent long-duration/seasonal storage and sector coupling; good reliability for off-grid	High cycle life, mechanical storage with low degradation; good for daily shifting/ancillary services	Mature and efficient for short (≈ 4 h) shifting; strong market evidence and tools for sizing/economics
Typical storage duration target	Minutes $\rightarrow$ days/seasonal (GES handles minutes–hours; H ₂ covers multi-day)	Multi-day/seasonal (electrolyzer + tank + fuel cell) [47]	Minutes–hours (depends on mass/height; project-scalable) [21]	≈ 4 h standard in utility PV+Battery designs; longer possible at higher cost/size [48]
Round-trip efficiency (RTE) trend	Hybridizes low H ₂ RTE with higher GES RTE to lift net system efficiency	Lower overall (electrolysis $\times$ FC), typically below batteries; RD&D aims to raise RTE [49]	Mechanical losses manageable; RTE competitive for short durations (empirical/analytical studies) [50]	High vs. H ₂ ; DC-coupled PV+Battery assumes ~ 85 – 87% charge/discharge pathway efficiencies [51]
Degradation & cycling	GES takes high-frequency cycles; H ₂ used less often (life extension for FC/electrolyzer)	FC/electrolyzer cycling and stack life are cost/risk drivers; careful dispatch needed [49]	Low wear vs. batteries; mainly mechanical maintenance; long service life reported [52]	Battery degradation & replacements materially affect sizing (often 2–3 replacements over life in studies) [53]
COE/LCOE from recent studies (context-specific, not directly comparable)	LCOE = 0.23 €/kWhFor LPSP=0%	COE ≈ 0.443 \$/kWh (NPC \$138k); sensitivity shows strong resource/capex dependence [49]	The cost of energy of RE-GES ranges between 0.3 €/kWh and 0.018 €/kWh.[46]	Grid-connected PV–WT–Li-ion: sizing tools/joint optimization show economic viability; battery replacements drive cost; LCOE between 18.73 and 22.26 euro/MWh [54]
System reliability notes	Enables near-zero LPSP at lower curtailment (GES smooths intra-day; H ₂ covers extended deficits)	Achieves low LPSP but at higher COE if H ₂ components are expensive/scarc [49]	High reliability for intra-day; multi-day events require oversizing [46]	Very effective intra-day; reliability over multi-day deficits requires large/expensive batteries or a second store [48]
Scalability & siting	Modular; H ₂ tanks scale well; GES needs suitable structure/elevation but is site-flexible vs. pumped hydro	Mature toolchains (HOMER, etc.); H ₂ BOS, safety, and FC lifetime are key constraints [7].	Site-flexible mechanical builds; growing evidence for grid support/ancillary value [55]	Widely deployed; strong cost curves and policy support; best fit for short-duration markets[56].

and round-trip efficiency [50]. GES maintenance focuses on seal inspection/replacement, pump-turbine overhauls, and hydraulic checks; the long mechanical life and minimal degradation complement the limited cycling of the FC seen in our results [57].

The optimized sizes of electrolyzer ≈ 790 MW, fuel cell ≈ 650 MW and H_2 tank ≈ 260 t are inherently modular: commercial alkaline electrolyzers are deployed as 10–20 MW skids aggregated to GW scale, and fuel-cell blocks can be paralleled in tens-of-MW increments [58]. Our base techno-economic model is agnostic to the storage type and enforces mass-inventory limits and pressure-agnostic throughput bounds at dispatch; detailed thermodynamics and compression work are noted as future engineering details. Hydrogen O&M includes periodic leak surveys, compressor maintenance, and sensor/calibration routines.

The 594 turbines (V164-9.5 MW) and ≈ 0.44 MW PV array in our case study are located in a resource-rich Moroccan setting; practical layouts depend on permitting, topography, and logistics. Micro-siting, wake management, and land/sea-use are standard wind-farm practice; the PV footprint is small relative to the wind array [59].

Key uncertainties of the proposed hybrid system include geotechnical risk for deep shafts, local water sourcing and treatment, and H_2 storage form-factor availability. We can mitigate these with site screening and staged Geotech, dedicated water-treatment units sized to the 139–153 m^3/h peak requirement, and storage modularization.

A practical rollout of the proposed hybrid system can follow three phases. Phase 1 establishes microgrid stability by deploying the initial PV–wind capacity, ~ 25 –40% of the electrolyzer/fuel cell units, and a single GES shaft. Phase 2 expands hydrogen infrastructure and storage, incrementally adding EL/FC blocks and H_2 tanks to achieve near-zero LPSP over an annual cycle. Phase 3 finalizes redundancy and optimization by commissioning parallel or uprated GES shafts, integrating spare units, and refining EMS constraints. Scalability is achieved through modular expansion: PV and wind scale with annual energy demand, electrolyzers with surplus capture, fuel cells with deficit coverage, GES rated power with ramp-rate requirements, and hydrogen mass inventory with the long-duration mismatch between resource availability and load.

7. Conclusion

As a sustainable substitute for an ultra-supercritical coal-fired power plant in Morocco, this study offers a thorough analysis of the simultaneous optimization and control of an off-grid hybrid system that includes photovoltaics (PV), wind turbines, hydrogen storage (electrolyzer (EL) and fuel cell (FC)), and a gravity energy storage unit. The system's size and operation are designed to provide high reliability, which is represented by the loss of power supply probability.

The analysis confirms that an integrated PV–Wind–Hydrogen–GES system can supply a continuous, reliable electricity output to match the load demand, even under 0% LPSP conditions. The optimal setup includes approximately 1000 PV modules, 594 wind turbines, a GES unit with a 5 m diameter and 714 m height, and significant hydrogen infrastructure (790 MW electrolyzer, 650 MW fuel cell (FC), and a 260 ton storage tank), yielding a levelized cost of electricity (LCOE) of 0.23 €/kWh. The results demonstrate how surplus wind and solar energy can be converted into hydrogen or stored gravitationally, then dispatched via the fuel cell and GES when renewable resources are insufficient. This flexibility is crucial to ensuring seamless coverage of the load despite significant variability in solar irradiance, wind speed, and consumer demand.

Key findings from the optimization include the importance of optimal sizing of renewable generation to guarantee full reliability and the consequential role of storage in capturing excess power for later use. Hydrogen production correlates strongly with renewable availability, peaking in periods of high solar and wind potential. Meanwhile, the GES frequently cycles between nearly full and fully discharged states, serving as an additional buffer. The fuel cell operates intermittently at high

power, particularly when wind and PV production dip, which illustrates how hydrogen storage complements variable renewables.

An increase in allowable LPSP would significantly reduce the levelized costs and installed capacity because of the trade-off between system reliability and economic optimization. The proposed hybrid system achieves an interesting lifetime cost (\$0.23/kWh) compared to coal power plants (\$0.18–0.34/kWh) while meeting emissions targets. This demonstrates the feasibility of replacing coal plants with renewable hybrid systems.

While the present study demonstrates the feasibility of a PV–WT– H_2 –GES hybrid to replace a 624 MW coal plant at LPSP = 0%, several limitations must be acknowledged. The dispatch was modeled at hourly resolution, which omits fast transient dynamics. Component degradation and maintenance cycles (e.g., fuel cell stack replacements, GES seal/turbine wear) were simplified to cost assumptions rather than explicitly simulated. Grid dynamics and interconnection constraints were not represented, as we assumed an islanded load profile. Hydrogen storage details such as compression energy, leakage rates, and process-safety engineering were simplified, although qualitative discussion of safety considerations has been added. Finally, experimental validation of full-scale GES and integrated hydrogen–GES systems remains outstanding. Future work should extend the modeling to sub-hour time steps, integrate component aging and O&M schedules, assess grid-coupling dynamics, and validate assumptions through pilot demonstrations. The proposed hybrid configuration offers a feasible, greener replacement for coal power plants. This would contribute to Morocco's energy security and decarbonization objectives. Future work could focus on refining cost assumptions, integrating demand-side management, and assessing long-term performance under evolving market and climatic conditions.

CRedit authorship contribution statement

Mohammed Sahab: Writing – review & editing, Writing – original draft, Visualization, Validation. **Anisa Emrani:** Writing – review & editing, Writing – original draft, Visualization, Validation. **Mohammad J. Sanjari:** Writing – review & editing, Writing – original draft, Visualization, Validation. **Jamil Abdelmajid:** Writing – review & editing, Writing – original draft, Visualization, Validation. **Asmae Berrada:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors are unable or have chosen not to specify which data has been used.

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